

Learning and Behavior

PSY/NSC 3320

Jean P. Bodet III, Ph.D.

Baylor University

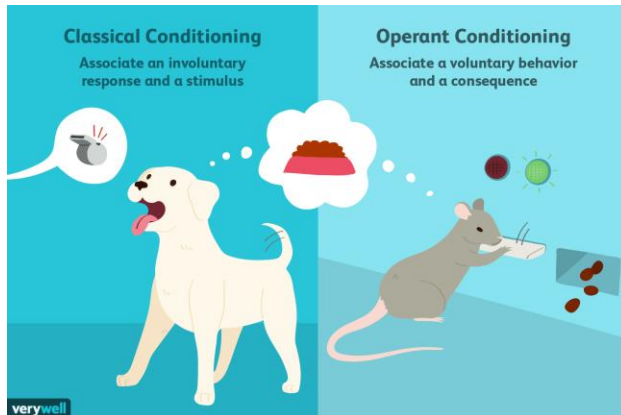
PREVIOUSLY ON
Lrng & Bhr...

Key Concepts

Respondent (Classical) Conditioning

Pairing two stimuli to each elicit a response

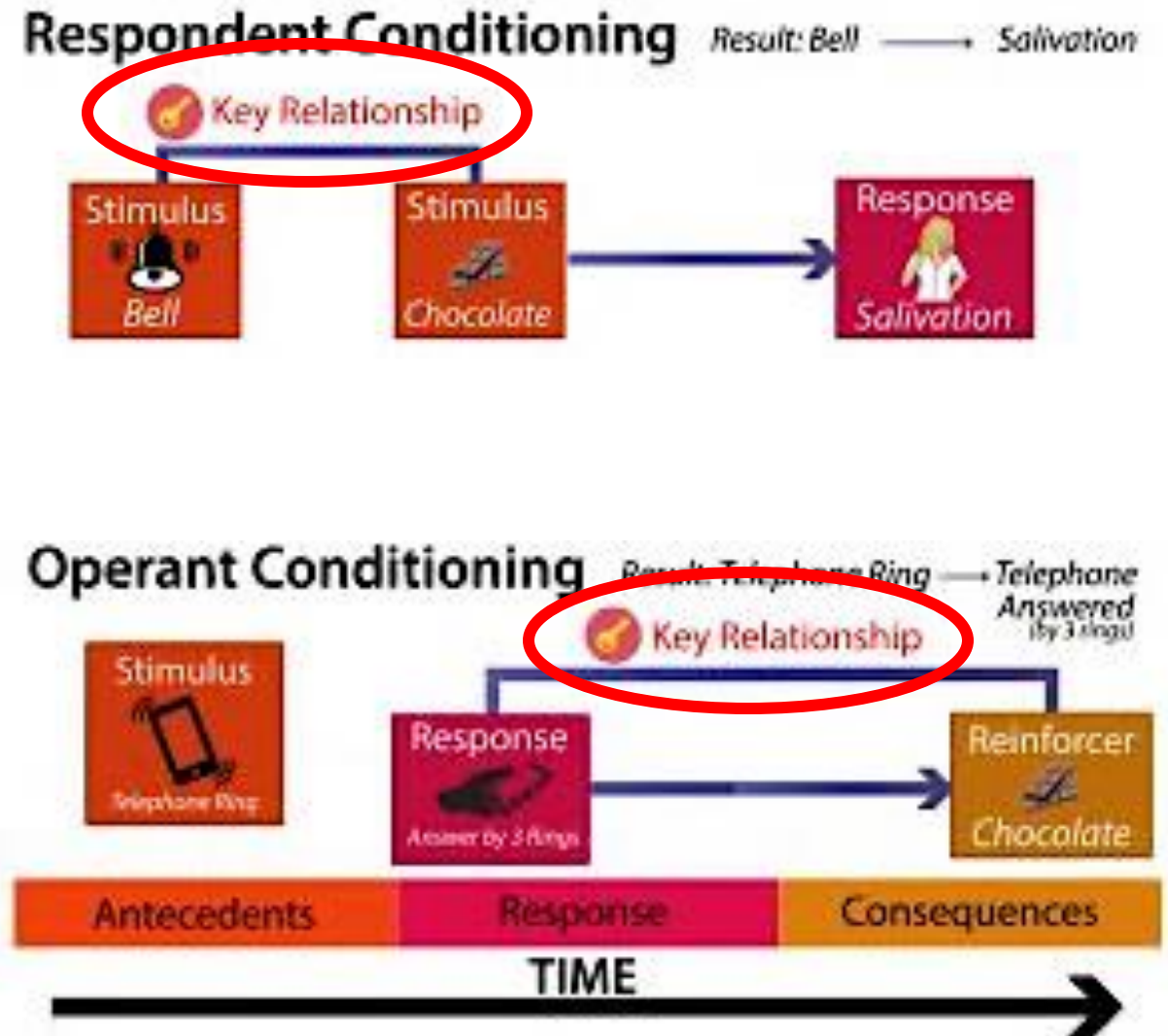
Food Salivate



Operant Conditioning

Pairing a voluntary response to a consequence

Press a bell → Food



Types of Antecedents

Discriminative Stimulus

(S^D)

An antecedent that signals the response will be reinforced

Discriminative Stimulus for Punishment

(S^{Dp})

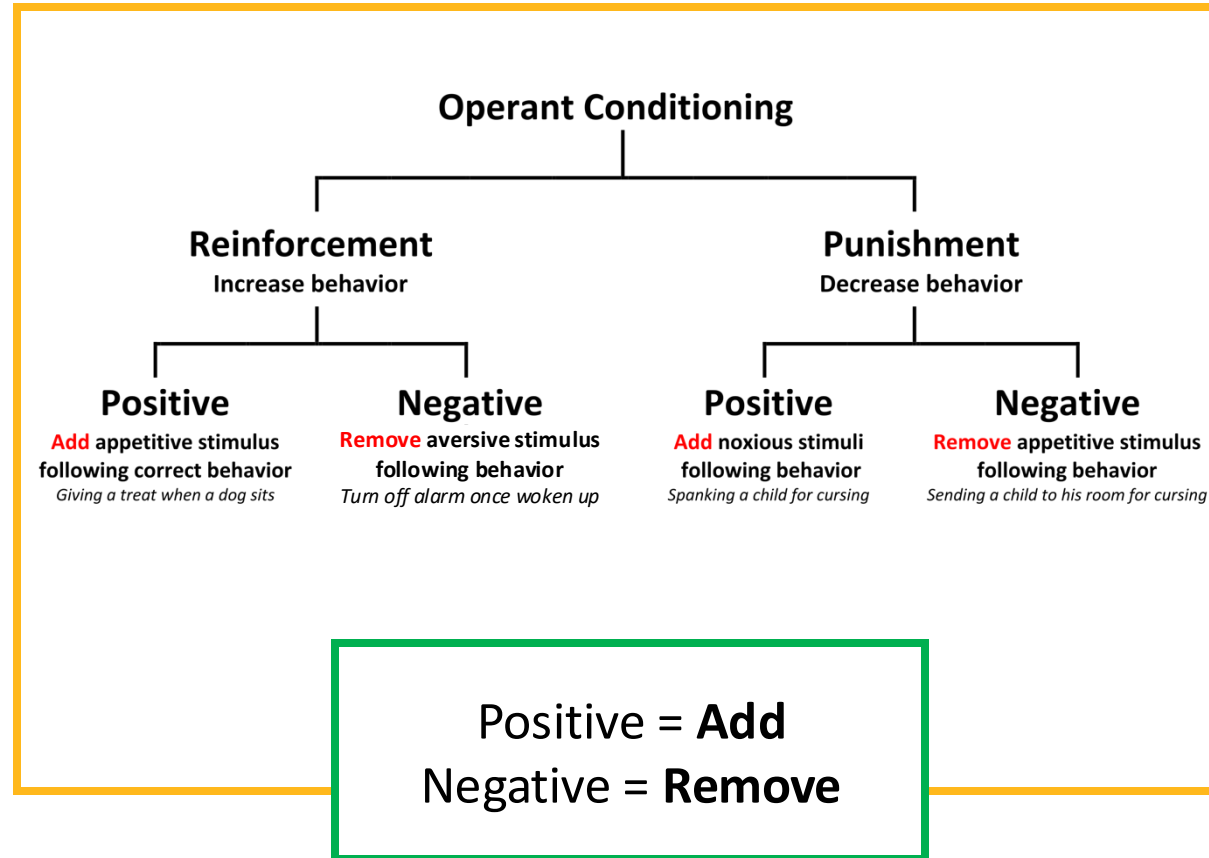
An antecedent that signals the response will be punished

Discriminative Stimulus for Extinction

(Extinction Stimulus, S^Δ)

An antecedent that signals the response will not be strengthened

Contingencies



Types of Consequence Stimuli

Immediate vs Delayed

Primary vs Secondary

Intrinsic vs Extrinsic

Natural vs Contrived

Operant Conditioning Pt 2

Reinforcement

Simple Schedules

- Continuous & Intermittent (4 Types)
- Fixed & Variable Duration
- DRH, DRL, & DRP Response Rates
- Fixed & Variable Time

Theories

- Drive Reduction Theory
- Premack Principle
- Response Deprivation Hypothesis

Simple Reinforcement Schedules

Schedule of Reinforcement

Response requirement that must be met
to obtain reinforcement (defined by experimenter)

For instance:

Food pellet every press

vs.

Food pellet every 10 presses

Simple Reinforcement Schedules

Simple Reinforcement Schedules:

- Continuous & Intermittent (4 Types)
- Fixed & Variable Duration
- DRH, DRL, & DRP Response Rates
- Fixed & Variable Time

Simple Reinforcement Schedules

- Continuous Reinforcement
 - Every response → reinforcement
 - *Use to establish new behavior*
- Intermittent Reinforcement
 - Some responses → reinforcement
 - *Use to keep new behavior*

Simple Reinforcement Schedules

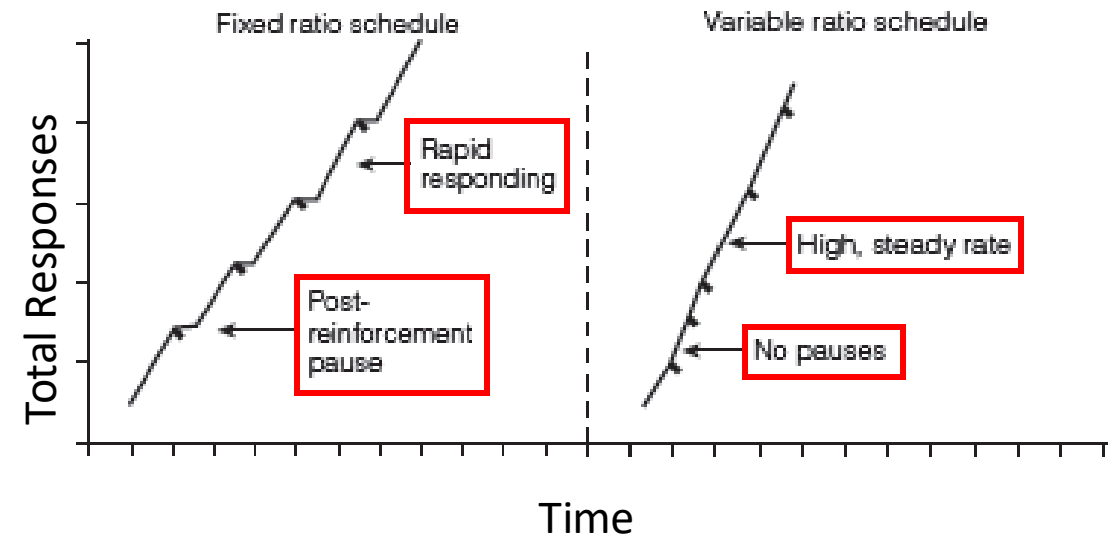
Intermittent (AKA “Partial”) Reinforcement

- Fixed Ratio
- Variable Ratio
- Fixed Interval
- Variable Interval

Simple Reinforcement Schedules

Intermittent Reinforcement

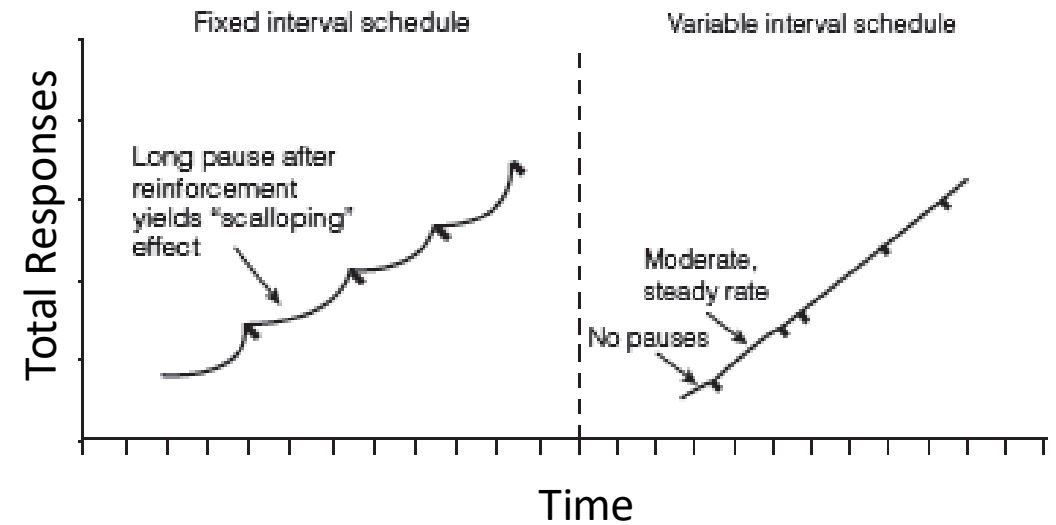
- **Fixed Ratio** – every X behaviors → reinf.
 - Rapid behaviors
 - Receive reward
 - Quick break
 - Rapid behaviors
- **Variable Ratio** – unpredictable (behaviors)
 - Rapid behaviors
 - Receive reward
 - NO break
 - Rapid behaviors



Simple Reinforcement Schedules

Intermittent Reinforcement

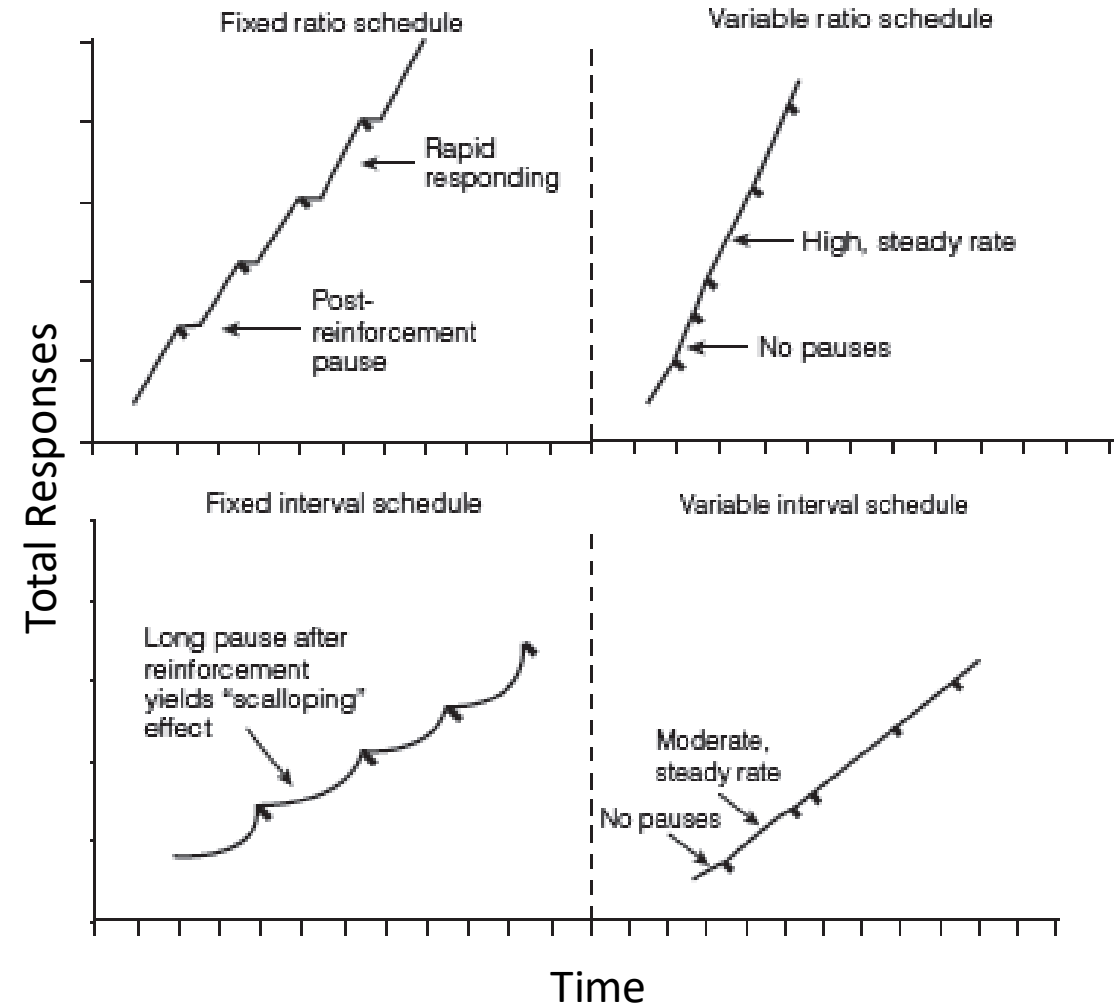
- **Fixed Interval** – every X sec. → reinf.
 - 1st response after a certain amt of time
 - “*scalloped*” responses
 - Reinforcement
 - Long Pause
 - Gradual increase until next reinforcement
- **Variable Interval** – unpredictable (time)
 - 1st response after a random amount of time
 - Moderate, steady responses
 - Little-to-no post-reinforcement pause



Simple Reinforcement Schedules

Intermittent Reinforcement

- Fixed Ratio – rapid; brief post-reinf. pause
- Variable Ratio – rapid; no pauses
- Fixed Interval – scalloped; long post-reinf. pauses
- Variable Interval – moderate; no pauses



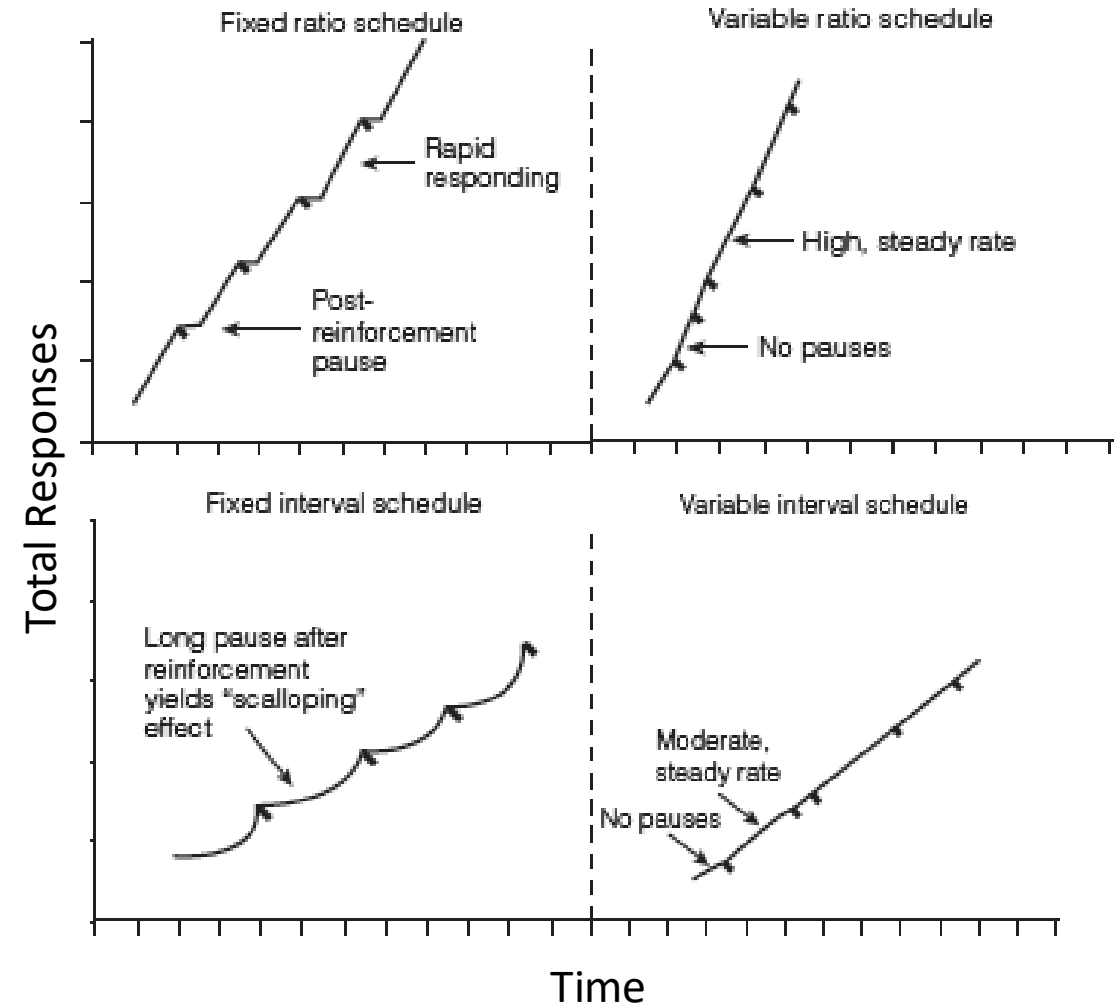
Simple Reinforcement Schedules

Intermittent Reinforcement

- Fixed Ratio – rapid; brief post-reinf. pause
- Variable Ratio – rapid; no pauses
- Fixed Interval – scalloped; long post-reinf. pauses
- Variable Interval – moderate; no pauses



Which of these is the most 'addictive'?



Simple Reinforcement Schedules

Simple Reinforcement Schedules:

- ~~Continuous & Intermittent (4 Types)~~
- Fixed & Variable Duration
- DRH, DRL, & DRP Response Rates
- Fixed & Variable Time

Simple Reinforcement Schedules

Duration Schedules

- Fixed Duration – run for 60s, get food
- Variable Duration – run for ?s, get food

Simple Reinforcement Schedules

Duration Schedules

- Fixed Duration – run for 60s, get food
- Variable Duration – run for ?s, get food
- Problem – no control over behavior intensity

Simple Reinforcement Schedules

Simple Reinforcement Schedules:

- ~~Continuous & Intermittent (4 Types)~~
- ~~Fixed & Variable Duration~~
- DRH, DRL, & DRP Response Rates
- Fixed & Variable Time

Simple Reinforcement Schedules

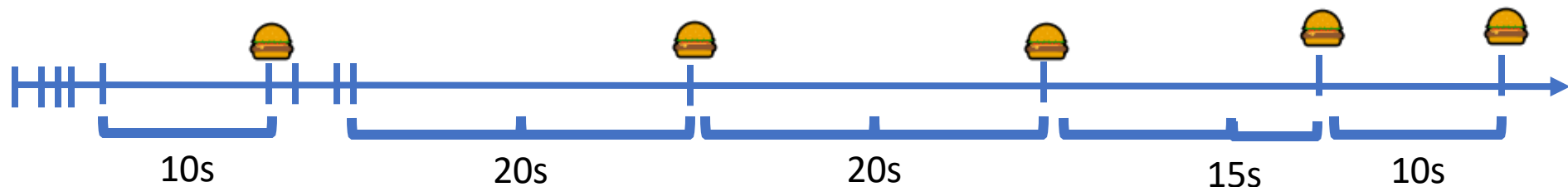
Response Rate Schedules

- Differential Reinforcement of High Rates (DRH)
- Differential Reinforcement of Low Rates (DRL)
- Differential Reinforcement of Paced Responding (DRP)

Simple Reinforcement Schedules

Response Rate Schedules

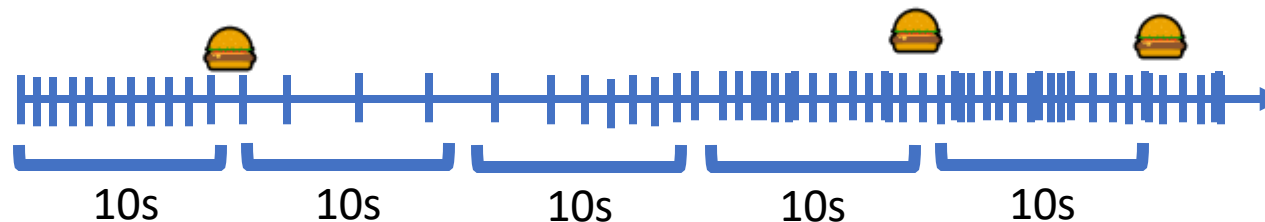
- Differential Reinforcement of High Rates (DRH)
 - >10 lever presses in 10s → food
 - Reinforces rate of response of at least this high intensity
- Differential Reinforcement of Low Rates (DRL)
 - >10s after last lever press, next lever press gets food
 - Reinforces rate of response of certain low intensity
- Differential Reinforcement of Paced Responding (DRP)
 - 10 lever presses with 1-2s between each
 - Reinforces specific range of rate of response



Simple Reinforcement Schedules

Response Rate Schedules

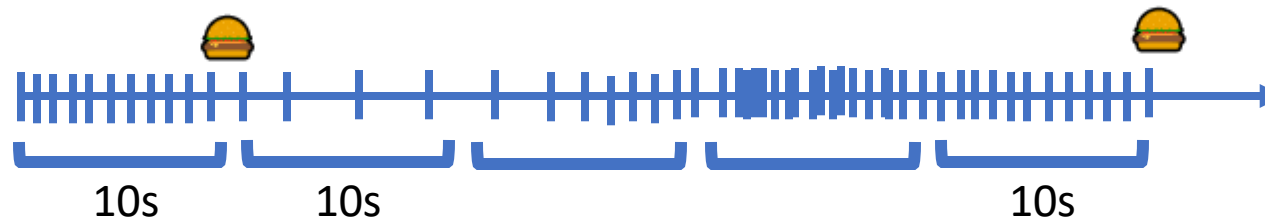
- **Differential Reinforcement of High Rates (DRH)**
 - >10 lever presses in 10s → food
 - Reinforces rate of response of at least this high intensity
- Differential Reinforcement of Low Rates (DRL)
 - >10s after last lever press, next lever press gets food
 - Reinforces rate of response of certain low intensity
- Differential Reinforcement of Paced Responding (DRP)
 - 10 lever presses with 1-2s between each
 - Reinforces specific range of rate of response



Simple Reinforcement Schedules

Response Rate Schedules

- Differential Reinforcement of High Rates (DRH)
 - >10 lever presses in 10s → food
 - Reinforces rate of response of at least this high intensity
- Differential Reinforcement of Low Rates (DRL)
 - >10s after last lever press, next lever press gets food
 - Reinforces rate of response of certain low intensity
- **Differential Reinforcement of Paced Responding (DRP)**
 - 10 lever presses with 1-2s between each
 - Reinforces specific range of rate of response



Simple Reinforcement Schedules

Simple Reinforcement Schedules:

- ~~Continuous & Intermittent (4 Types)~~
- ~~Fixed & Variable Duration~~
- ~~DRH, DRL, & DRP Response Rates~~
- Fixed & Variable Time

Simple Reinforcement Schedules

Non-Contingent Schedules:

- *Not contingent* on responses
- Fixed Time – food every 30s
- Variable Time – food randomly

Theories of Reinforcement

Drive Reduction Theory
Premack Principle
Response Deprivation Hypothesis

Theories of Reinforcement

Drive Reduction Theory

- Stimuli can reinforce a behavior if they reduce a physiological drive

Spin → Food (primary reinforcer, reduces hunger)
Spin more

Theories of Reinforcement

Drive Reduction Theory

- Stimuli can reinforce a behavior if they reduce a physiological drive

Work → Get money (secondary reinforcer, associated with reducing hunger/etc.)

Work more

Theories of Reinforcement

Drive Reduction Theory

- Stimuli (consequences) can reinforce a behavior if they reduce a physiological drive
- **Incentive Motivation** – when a stimulus reinforces a behavior because of some property of the stimulus

Go to Favorite Restaurant → Get Food (reduce hunger)

Restaurant depends on food you like – incentive motivation for one restaurant over the other

Theories of Reinforcement

Premack Principle

What makes for a good **reinforcer**? (and, thus, why are some reinforcers stronger than others?)

Theories of Reinforcement

Premack Principle

First, we must reframe the 'stimulus' (consequence) in an operant conditioning paradigm

Theories of Reinforcement

Premack Principle

Response → Stimulus

Behavior → Consequence

Behavior → **Reinforcing Behavior**

Lever Press → Eating (food)

Cleaning → Playing games (Xbox)

Theories of Reinforcement

Premack Principle

Behavior → Reinforcing Behavior

Lever Press [**LPB**] → Eating [**HPB**]

Cleaning [**LPB**] → Playing games [**HPB**]

Theories of Reinforcement

Premack Principle

High-probability behaviors can reinforce low-probability behaviors

Lever Press [**LPB**] → Eating [**HPB**]

Cleaning [**LPB**] → Playing games [**HPB**]

Theories of Reinforcement

Premack Principle

High-probability behaviors can reinforce low-probability behaviors

If not hungry...

Eating [**LPB**] → Playing games [**HPB**]

Theories of Reinforcement

Premack Principle

High-probability behaviors can reinforce low-probability behaviors

Consider what a LPB and a HPB is for any organism and you may build an effective reinforcement procedure

Theories of Reinforcement

Premack Principle

What makes for a good reinforcer? (and, thus, why are some reinforcers stronger than others?)

HPB make good reinforcers

The higher the probability of the reinforcer compared to the behavior of interest, the stronger the reinforcement

Theories of Reinforcement

Response Deprivation Hypothesis

Builds on same principle as Premack – framing
reinforcers as behaviors

Theories of Reinforcement

Response Deprivation Hypothesis

Behaviors can be reinforcers when:

- 1) Access to that behavior is restricted
- 2) The behavior's frequency thus falls below preferred frequency

Theories of Reinforcement

Response Deprivation Hypothesis

Eating has certain probability of occurring

Limit access to food

Eating falls below preferred frequency

Probability of eating rises

Increases potency of eating as reinforcer

Theories of Reinforcement

Response Deprivation Hypothesis

Cleaning has certain probability of occurring

Limit access to cleaning tools

Cleaning falls below preferred frequency

Probability of cleaning rises

Cleaning can now be strong reinforcer

Extinction

Extinguished = fully extinct
Partially extinguished = some behavior still

Extinction

Extinction

Side Effects
Resistance
Spontaneous Recovery

Extinction

Side Effects

Extinction Burst – sudden increase in behavior

Increase in Variability – variance in performance of behavior

Emotional Behavior (typically, frustration)

Aggression

Resurgence – reappearance of previously reinforced behaviors

Depression

Extinction

Resistance to Extinction

Influenced by...

Schedule of Reinforcement – intermittent is more resistant than continuous

History of Reinforcement – more reinforcements leads to more resistance

Magnitude of Reinforcer – bigger reinforcers lead to more resistance

Degree of Deprivation – hungry rat is more likely to keep pressing

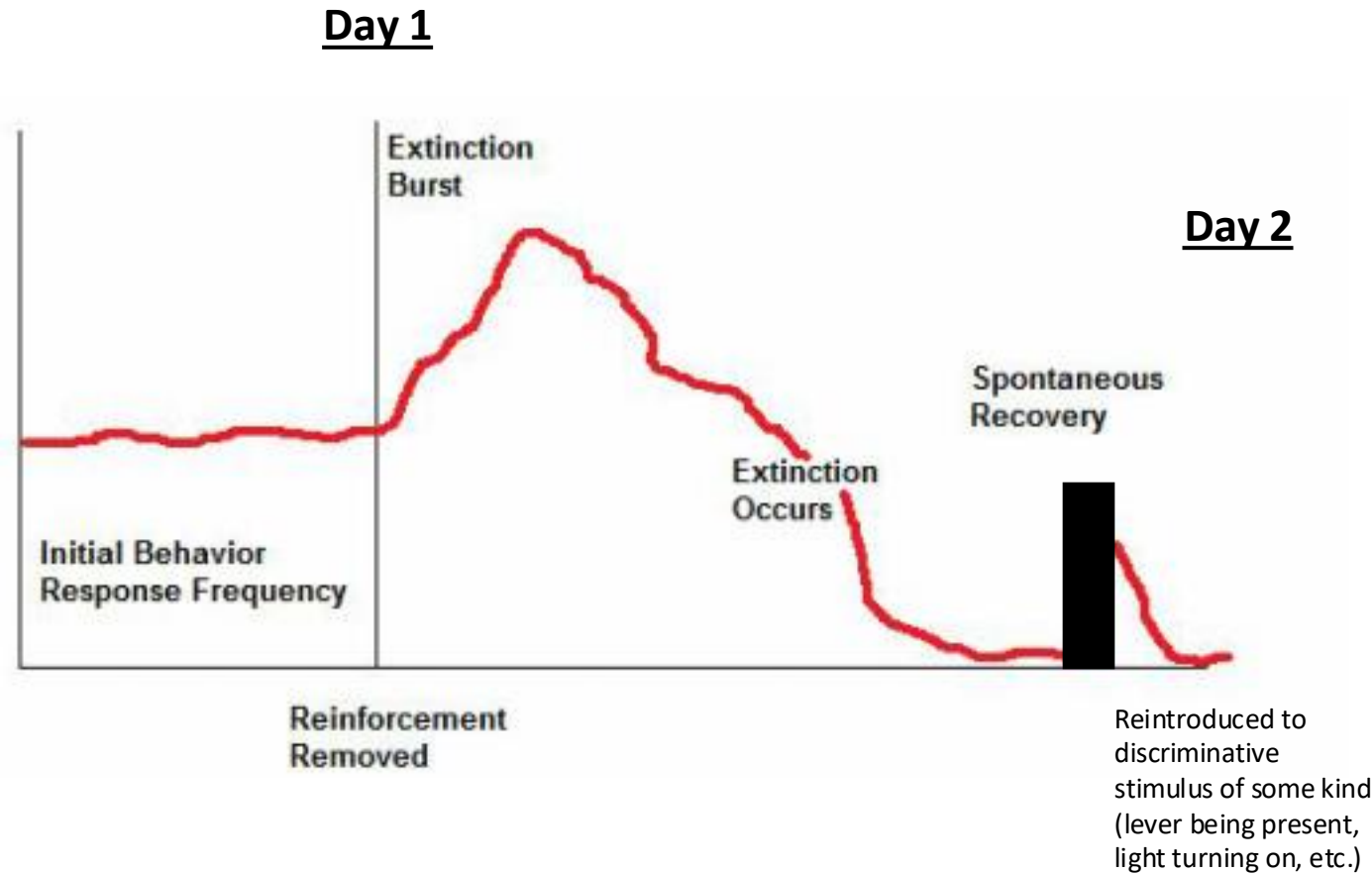
Previous Experience with Extinction – more past extinctions → faster

Discriminative Stimulus for Extinction – cued onset of extinction → faster

Extinction

Spontaneous Recovery

reappearance of extinguished response, even without reinforcement, after a break



Questions?